Program 1

Problem Statement
Define an abstract 'Payment' base class with a pure virtual function and concrete derived payment classes that override it, then dispatch the correct payment computation from a common base-class pointer.

Source Code:

```
#include <iostream>
#include <string>
using namespace std;

class Payment {
protected:
string name, mobile_no, address, confirmation;
double amount;

public:
Payment(string n, string m, string add, double amt) {
name = n;
mobile_no = m;
address = add;
amount = amt;
confirmation = "No";
}

void showInfo() {
cout << "Name: " << name << endl;
cout << "Mobile Number: " << mobile_no << endl;
cout << "Address: " << address << endl;
if (confirmation == "Yes")
cout << "Total Amount: " << amount << endl;
else
cout << "Invalid Payment Method" << endl;
}

virtual void processPayment() = 0;
};

class CashonDelivery : public Payment {
public:
CashonDelivery(string n, string m, string add, double amt)
: Payment(n, m, add, amt) {}

void processPayment() {
double charge;
if (address == "Dhaka")
charge = 80;
else
charge = 150;

confirmation = "Yes";
```

```
cout << "Delivery charge: " << charge << endl;
cout << "Pay method: Cash on Delivery" << endl;
cout << "Final Total: " << amount + charge << endl;
}
};

class MobileBanking : public Payment {
string provider;
public:
MobileBanking(string n, string m, string add, double amt, string pvdr)
: Payment(n, m, add, amt) {
provider = pvdr;
}

void processPayment() {
double fee = 0;
if (provider == "bkash")
fee = 20;
else if (provider == "rocket")
fee = 15;
else if (provider == "nagad")
fee = 12;

confirmation = "Yes";
cout << "Pay method: " << provider << endl;
cout << "Fee: " << fee << endl;
cout << "Final Total: " << amount + fee << endl;
}
};

class CardPayment : public Payment {
public:
CardPayment(string n, string m, string add, double amt)
: Payment(n, m, add, amt) {}

void processPayment() {
double discount = (amount / 100) * 10;
confirmation = "Yes";
cout << "Pay method: Card Pay" << endl;
cout << "Discount: " << discount << endl;
cout << "Final Total: " << amount - discount << endl;
}
};

int main() {
Payment *p;

CashonDelivery a("Shamtha", "01711111111", "Natore", 1000);
MobileBanking b("Saima", "01855555555", "Rangpur", 3000, "bkash");
CardPayment c("Sajada", "01987654367", "Dhaka", 5000);
```

```
p = &a;
p→processPayment();
p→showInfo();

cout << endl;

p = &b;
p→processPayment();
p→showInfo();

cout << endl;

p = &c;
p→processPayment();
p→showInfo();

return 0;
}
```

Input:
(no input)

Output:

